Toeplitz operators and representation theory

Present:
Roberto García Antonio
CIMAT

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some historical background

1. The classical study takes place in Hardy spaces on the upper half-plane.
2. Toeplitz algebras appear in the Berezin-Toeplitz quantization, or in solid state physics[AM20] . This is another approach to quantization which produces explicit operators from classical observables

Toeplitz operators

Let (X, Σ, μ) be a measure space, $\mathcal{H}$ be a closed subspace of $L^2(X, \mu)$, and $P : L^2(X, \mu) \to \mathcal{H}$ denote the orthogonal projection onto this subspace.
For $\varphi \in L^\infty(X, \mu)$ (the symbol), the Toeplitz operator $T_\varphi : \mathcal{H} \to \mathcal{H}$ with symbol φ is defined by

$$T_\varphi f = P(\varphi f), \quad \text{for all } f \in \mathcal{H}.$$

Bergman spaces

Let D be a bounded symmetric domain. The weighted Bergman space $\mathcal{H}^2_\lambda(D)$ is the closed subspace of $L^2(D, \mu_\lambda)$ consisting of holomorphic functions. Where

$$\mu_\lambda(z) = c_\lambda k_D(z,z)^{1-\frac{\lambda}{p}} dz$$

In the case of $D = \mathbb{D}$ the measure reduces to

$$\mu_\lambda(z) = (\lambda+1)(1-|z|^2)^\lambda$$

The Existence Problem of Commutative C*-Algebras on Bergman Spaces

For a given symbol $\varphi \in L^\infty(D)$, the Toeplitz operator $T_\varphi^{(\lambda)}$ is defined as

$$T_\varphi^{(\lambda)} = B_\lambda \circ M_\varphi$$

In general, Toeplitz operators do not commute

$$T_\varphi^{(\lambda)} T_\psi^{(\lambda)} \neq T_\psi^{(\lambda)} T_\varphi^{(\lambda)}.$$

An interesting problem is to find an abelian C*-algebra generated by Toeplitz operators.

The Unit Ball $\mathbb{B}^n = \mathrm{SU}(n,1)/\mathrm{S}(\mathrm{U}(n)\times\mathrm{U}(1))$ case

For the case of the disk $\mathbb{D}$ the prove is solved and have the next families of conmutative C^* algebras:

1. Elliptic-U(1)
2. Hyperbolic-$\mathrm{SO}_0(1,1)$
3. Parabolic-N

Representation theory

A representation of a locally compact group G over a Hilbert space $\mathcal{H}$ is just a topological group homomorphism

$$\pi : \mathrm{G} \to \mathrm{U}(\mathcal{H})$$

where $\mathrm{U}(\mathcal{H})$ is endowed with the strong operator topology.

Interwiner operator

For two representations (π_1, V_1) and (π_2, V_2) of a group G, an intertwiner operator is a linear map $\mathrm{T} : \mathrm{V}_1 \to \mathrm{V}_2$ satisfying:

$$\mathrm{T}\pi_1(\mathrm{g}) = \pi_2(\mathrm{g})\mathrm{T} \quad \text{for all } \mathrm{g} \in \mathrm{G}$$

The set of operators on $\mathcal{H}$ that intertwine the representationπ is denoted by $\mathrm{End}_{\mathrm{G}}(\mathcal{H})$

Multiplicity free representation

A representation π of G is multiplicity-free if, for every nontrivial decomposition $\pi = \pi_1 \oplus \pi_2$, the representations π_1 and π_2 are disjoint

Example

Any Gelfand pair

Type I Lie group

The representation of G is type I, or of discrete type, if it is quasiequivalent to a multiplicity-free representation.

A topological group G is type I if all its representations are type I.

Example

Compact Lie groups

Holomorphic discrete series: The bridge

Let us consider the bounded symmetric domain $\mathrm{D} = \mathrm{G}/\mathrm{K}$ and the weighted Bergman space $\mathcal{H}^2_\lambda(\mathrm{D})$ for $\lambda > \mathrm{p} - 1$. The holomorphic discrete series is a representation of the universal covering group $\tilde{\mathrm{G}}$. The group action is given by:

$$\pi_\lambda(\mathrm{g})\mathrm{f}(\mathrm{z}) = \mathrm{j}(\mathrm{g}^{-1}, \mathrm{z})^{\frac{\lambda}{\mathrm{p}}} \mathrm{f}(\mathrm{g}^{-1}\mathrm{z})$$

where $\mathrm{j}(\mathrm{g}, \mathrm{z})$ is the complex Jacobian.

Intertwining Operators and Commutativity

The connection between Toeplitz operators and representation theory is established by the following equivalence [DÓQ15]:

Lemma (Intertwining Condition)

Let $\tilde{\mathrm{H}}$ be a subgroup of $\tilde{\mathrm{G}}$ and $\varphi \in \mathrm{L}^\infty(\mathrm{D})$. The Toeplitz operator $\mathrm{T}_\varphi^{(\lambda)}$ intertwines the restriction $\pi_\lambda|_{\tilde{\mathrm{H}}}$ if and only if its symbol φ is $\tilde{\mathrm{H}}$-invariant.

Main Theorem (Dawson, Ólafsson, Quiroga-Barranco, 2015)

Let H be a closed subgroup of G and $\mathcal{A}^{\mathrm{H}}$ the subspace of H-invariant bounded symbols on D. If the algebra of intertwining operators $\mathrm{End}_{\tilde{\mathrm{H}}}(\mathcal{H}^2_\lambda(\mathrm{D}))$ is commutative, then the generated C*-algebra $\mathcal{T}^{(\lambda)}(\mathcal{A}^{\mathrm{H}})$ is commutative.
In particular, this holds if H is a type I group and the restriction $\pi_\lambda|_{\tilde{\mathrm{H}}}$ is multiplicity-free.

By applying this theorem to K, the maximal compact subgroup of G, we obtain a generalization of the $\mathbb{B}^n$ case.

Corollary

Let $\mathcal{A}^K$ be the set of bounded symbols that are K-invariant. Then, the C*-algebra $\mathcal{T}^{(\lambda)}(\mathcal{A}^K)$ is commutative on the weighted Bergman space $\mathcal{H}^2_\lambda(D)$ for every $\lambda > p - 1$.

Commutative subgroups of biholomorphisms of $\mathbb{B}^n$

There are exactly $n+3$ essentially different types of such subgroups [QV07]:

- Quasi-elliptic: Isomorphic to $\mathbb{T}^n$, acting on the unit ball $\mathbb{B}^n$.
- Quasi-parabolic: Isomorphic to $\mathbb{T}^{n-1} \times \mathbb{R}$, acting on the Siegel domain D_n.
- Quasi-hyperbolic: Isomorphic to $\mathbb{T}^{n-1} \times \mathbb{R}_+$, acting on D_n.
- Nilpotent: Isomorphic to $\mathbb{R}^{n-1} \times \mathbb{R}$, acting on D_n.
- Quasi-nilpotent: Isomorphic to $\mathbb{T}^k \times \mathbb{R}^{n-k-1} \times \mathbb{R}$ (for $0 < k < n-1$), acting on D_n.
- Invariant under the action of $SO(n,1) \subset SU(n,1)$

Thanks for your attention :D

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